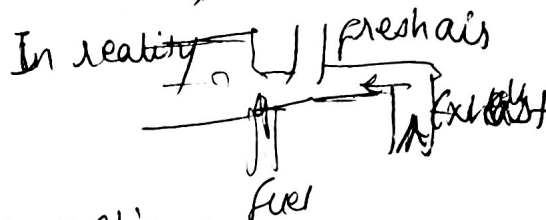


## Air-standard assumptions

- Working fluid is air (ideal gas) & circulates in a closed loop.
- All-cycle processes are internally reversible.  $\rightarrow$  not true
- Combustion processes replaced by heat addition process from external source.
- Exhaust process replaced by heat rejection  $\rightarrow$  to cool down

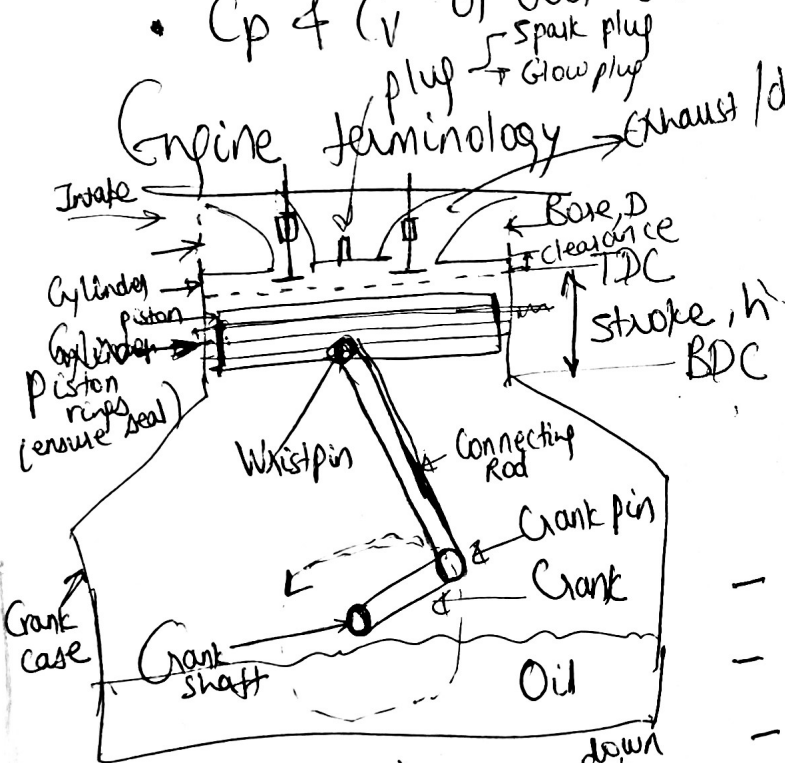


## (Cold) Air-standard Assumptions

- $C_p$  &  $C_v$  of air are taken to be @  $25^\circ\text{C}$  & constant

SI - Spark ignition

CI - Compression ignition



Oil ring - wipes the oil from the cylinder & itself.

- MEP - mean effective pressure  
 $W_{net} = MEP \times \text{piston area} \times \text{stroke}$

$$MEP = \frac{W_{net}}{V_{max} - V_{min}}$$

How much efficiency are compressing in cylinder

clearance volume  
 The volume of gas of cylinder at TDC  
 $\rightarrow$  Volume of gas in cylinder when piston is at TDC  
 Which enters

- TDC - cylinder at highest pt.
- BDC - " " lowest "
- Stroke - the distance piston moves up & down

- Bore - diameter of cylinder
- Displacement volume: volume displaced by the piston as it goes from TDC to BDC

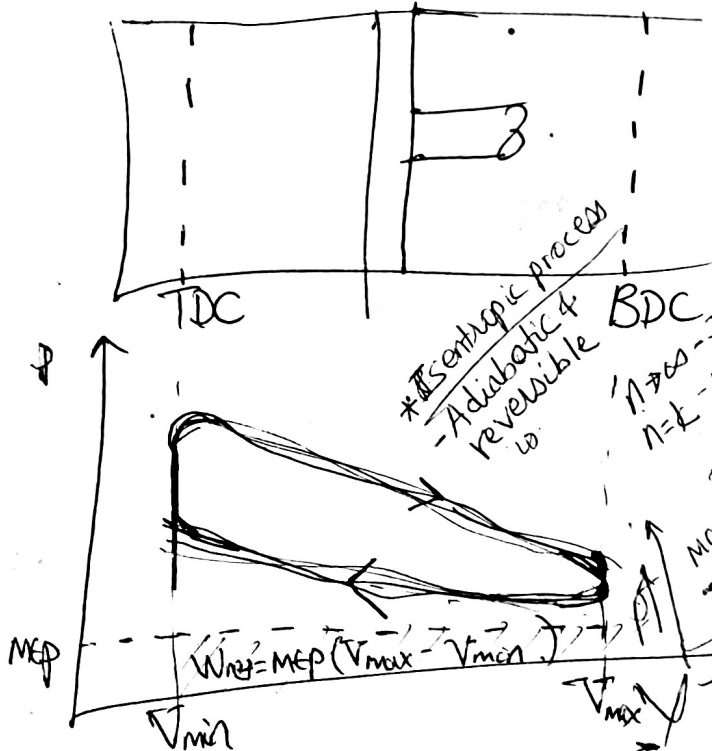
- Compression ratio  $r = \frac{V_{max}}{V_{min}} = \frac{V_{BDC}}{V_{TDC}}$

Heat work in motion  $\rightarrow$  change of state of system & surroundings  $\rightarrow$  path function

$$W = \int_1^2 P dV$$

From BDR  $\rightarrow$  AA  
- not state functions

$$MEP = \frac{W_{net}}{V_{max} - V_{min}} = \frac{W_{net}}{V_{max} - V_{min}}$$

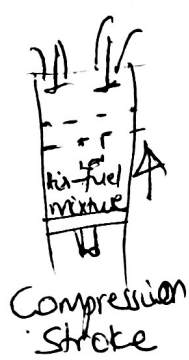


\* Isentropic process  
- Adiabatic & reversible

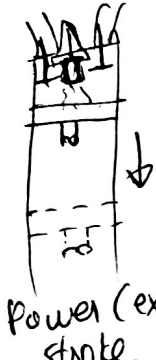
Isometric process  
- Isobaric process  
- Isentropic process  
- Isochoric process

$n = k$  - Isentropic process  
- Monatomic  
- Diatomic  
- Triatomic

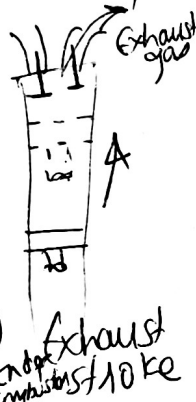
State function  
not depend on how it reaches state  $\delta, U, H, S$   
Path function  
Depend on route taken b/w two states



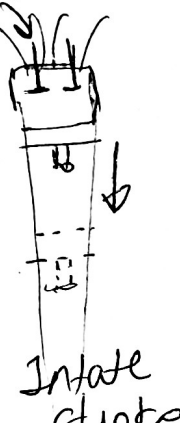
Compression stroke



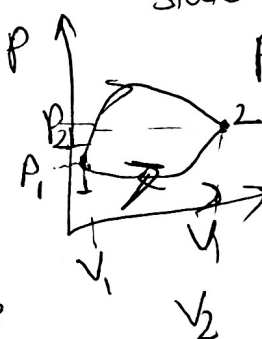
Power (expansion) stroke



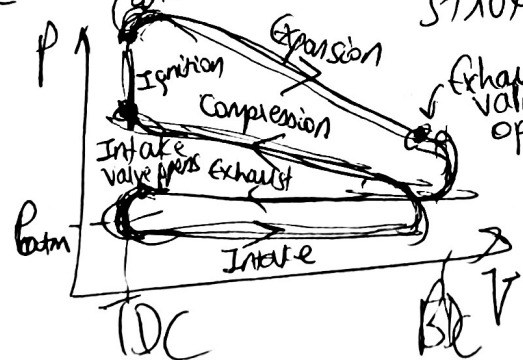
Exhaust stroke



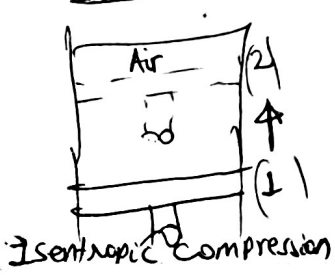
Intake stroke



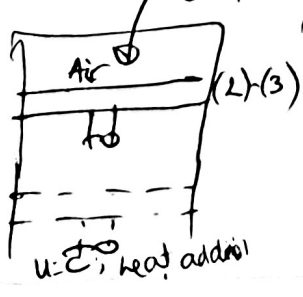
Actual



Ideal



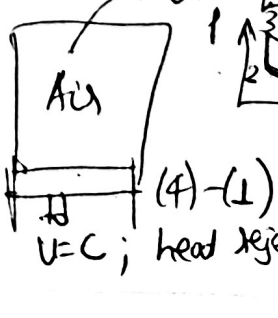
Isentropic compression



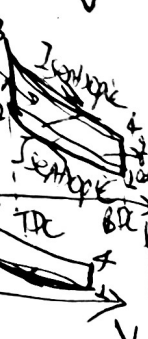
Isentropic expansion



Isentropic compression



Isentropic expansion



Isentropic compression

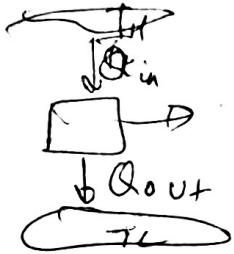
# Ideal Otto cycle

The mechanism may go through a cycle but the working fluid does not.

$\Delta K.E., \Delta P.E. \approx 0$  since whole engine is not moving macroscopically  
 $v = \text{not } \frac{1}{2} \frac{dK}{dt}$

First law 2-3  $\dot{Q}_{in} - \dot{W} = \Delta U = C_v(T_3 - T_2)$

4-1:  $-\dot{Q}_{out} - \dot{W} = \Delta U = C_v(T_1 - T_4)$   
 (heat is leaving)



$$\eta_{th} = \frac{W_{net}}{\dot{Q}_{in}} = \frac{\dot{Q}_{in} - \dot{Q}_{out}}{\dot{Q}_{in}}$$

$$= 1 - \frac{\dot{Q}_{out}}{\dot{Q}_{in}}$$

$$= 1 - \frac{\dot{Q}_{out}}{\dot{Q}_{in}} = 1 + \frac{C_v(T_1 - T_4)}{C_v(T_3 - T_2)}$$

$$= 1 + \frac{(T_1 - T_4)}{(T_3 - T_2)}$$

$$\eta_{th, Otto} = 1 - \frac{T_1 \left( \frac{T_4}{T_1} - 1 \right)}{T_2 \left( \frac{T_3}{T_2} - 1 \right)}$$

Using cold-air standard assumptions  
 1-2 & 3-4 are isentropic (reversible & adiabatic): thus  
 we can use the isentropic relations with  $V_2 = V_3$  &  $V_4 = V_1$

$$\frac{T_1}{T_2} = \left( \frac{V_2}{V_1} \right)^{k-1} = \left( \frac{V_3}{V_4} \right)^{k-1} = \frac{T_4}{T_3}$$

After substituting:  $\eta_{th, Otto} = 1 - \frac{1}{r^{k-1}}$

$r$  - compression ratio according to design & function

$$r = \frac{V_{max}}{V_{min}} = \frac{V_1}{V_2} = \frac{T_1}{T_2}$$

$$\& k = \frac{C_p}{C_v}$$

Typical compression ratios are b/n 11.5:1 & 22:1

↳ Bieng Diesel's usually operate at higher compression ratios, the overall thermal efficiency is usually higher than the Otto cycle.

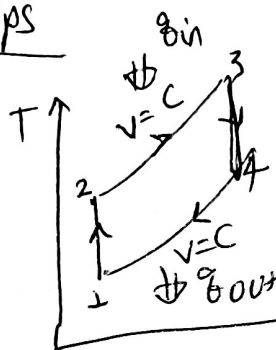
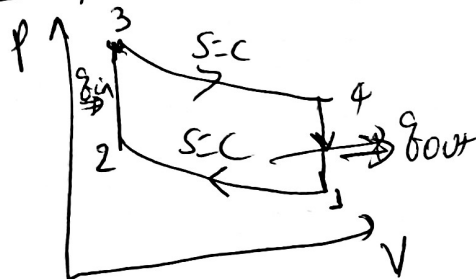
↳ louder engine  $\rightarrow$  higher pressure combustion <sup>high efficient</sup>  
Quieter engine  $\rightarrow$  less efficient

Otto & Diesel cycles

not externally reversible due to heat transfer through finite temp. difference  $\Delta T$   
 $\downarrow$  reduce efficiency

Problem solving tips

Otto cycle



know  $r = \frac{V_1}{V_2}$

, use specific volume to jump from (1-2) to (3-4)

$T_1 \rightarrow V_{r1}$

$$\frac{V_{r1}}{V_{r2}} = \frac{V_1}{V_2} = r \Rightarrow V_{r2} \Rightarrow T_2$$

$$T_1 = 60^\circ\text{C} + 273 = 333\text{ K}$$

Ideal gas law to get P:

$$P = \frac{RT}{V} \Rightarrow R = \frac{PV}{T}$$

$$\frac{P_2 V_2}{T_2} = \frac{P_1 V_1}{T_1} \quad V_{r1} = 330 - 489.4$$

$$333 - ?$$

$\rightarrow (2-3), (4-1) \Rightarrow V=C$

2-3 involves heat addition, const V,  $dv=0$

$$q - w = U_3 - U_2$$

$$q - w_{net} + w_b$$

$$\oint p dv = 0; v=0$$

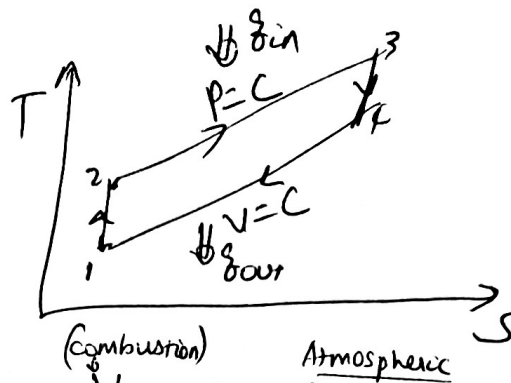
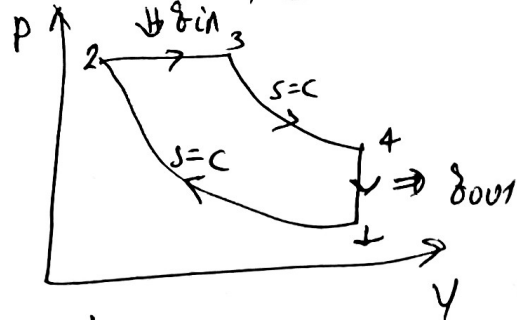
$$\Rightarrow q_{in} = U_3 - U_2$$

$$\text{Similarly } q_{out} = U_4 - U_1 \Rightarrow w_{net} = q_{in} - q_{out}$$

$$\frac{333 - 330}{340 - 330} = \frac{? - 489.4}{454.1 - 489.4}$$

$$\frac{3}{10} = \frac{?}{35.3}$$

# Diesel cycle



knowing  $r = \frac{v_1}{v_2}$  &  $r_c = \frac{v_3}{v_2}$  &  $P_1, T_1$

(1→2), (3→4) isentropic processes

Relative p & v  
from table

$$\left( \frac{v_{r2}}{v_1} \right)_{s=\text{constant}} = \left( \frac{v_2}{v_1} \right) = \frac{1}{r}$$

$$\left( \frac{p_{r2}}{p_1} \right)_{s=\text{constant}} = \frac{p_2}{p_1}$$

$$p_3 = p_2$$

From ideal gas law

$$\frac{p_3 v_3}{T_3} = \frac{p_2 v_2}{T_2}$$

1st law:  $\frac{2-3}{p=c}$  for fixed mass

$$q - w = u_3 - u_2$$

$$w_b = \int_2^3 p dv = p v \Big|_2^3; dv \neq 0$$

$$\therefore q_{in} = h_3 - h_2 \quad \text{includes boundary work when volume changes}$$

For  $q_{out}$ :

$$q - w = u_4 - u_1 \quad \text{when } v=c$$

$$q_{out} = u_4 - u_1$$

$$W_{net} = q_{in} - q_{out}$$

9-82  
9-83  
9-1084 Day  
9-143  
G

# Otto cycle #9-41

A four-cylinder, four stroke, 2.2-L gasoline engine operates on the Otto cycle with a compression ratio of 10. The air is @ 100kpa & 60°C at the beginning of the compression process, & the max. pressure in the cycle is 8 Mpa. The compression & expansion may be modeled as polytropic with a polytropic constant of 1.3. Using constant specific heats @ 850K, determine:-

- the temp. @ the end of the expansion process,  $T_4$
- the net work output & the thermal efficiency,  $W_{net,out}$ ,  $\eta_{th}$
- the mean effective pressure, MEP
- the engine speed for a net power output of 70kW,  $\dot{Q}$ ,  $n$
- the specific fuel consumption, in g/kWh, defined as the ratio of the mass of the fuel consumed to the net work produced. The air-fuel ratio, defined as the amount of air divided by the amount of fuel intake, is 16

## Solution

### Given

Four cylinder / four stroke  
2.2 L gasoline engine

$r = 10$   
state 1 AF = 16

$P_1 = 100 \text{ kPa}$

$T_1 = 60^\circ \text{C}$

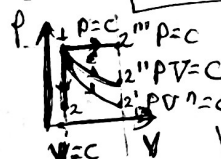
$P_3 = 8 \text{ MPa}$

Compression & Expansion } Polytropic  
 $n = 1.3$

- 1-2'''  $\Rightarrow$
- 1-2''  $\Rightarrow$
- 1-2'  $\Rightarrow$
- 1-2  $\Rightarrow$

### Required

- $T_4$  from
- $W_{net,out}$  &  $\eta_{th}$
- MEP
- $n$  @ 70kW
- Fuel



$\delta W = P dv$

Constant sp. heats @ 850K:

$C_p = 1.110 \text{ kJ/kg.K}$

$C_v = 0.823 \text{ kJ/kg.K}$

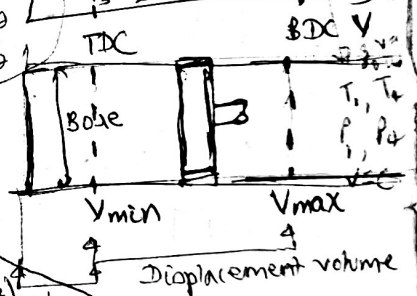
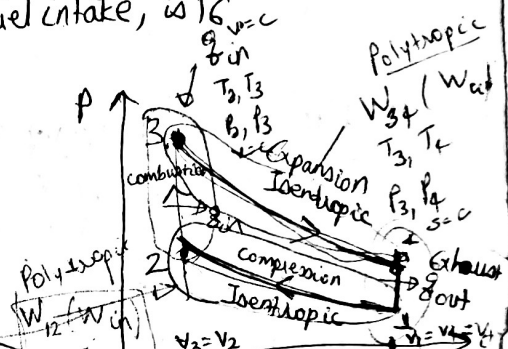
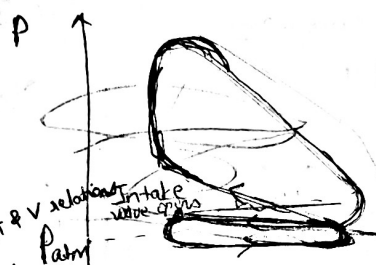
$R = 0.287 \text{ kJ/kg.K}$

$k = 1.349$

Table A-26

### Assumptions:

- Air-standard assumptions are applicable.
- $\Delta KE, \Delta PE$  are negligible.
- Air is ideal gas with constant specific heats.



Isentropic (reversible & adiabatic) process:  $k$

Polytropic process;  $n$  (for polytropic path)

Expansion/compression

Rapid process - Isentropic =  $Pv^k = C$   
to that allowed - Isentropic =  $Pv^k = C$   
Air & Fuel - Polytropic  $\Rightarrow$  somewhere in b/w  
adiabatic  
Isentropic process

$\frac{50}{100} = \frac{x - 1.099}{1.121 - 1.099}$   
 $C_{p@850} = x = 1.11$

$C = P v^n = P_1 v_1^n = P_2 v_2^n$

Ideal gas:  $P v = R T$

@ 800K  $\frac{C_p}{R} = 1.099$   $\frac{C_v}{R} = 0.812$   $\frac{P}{T} = \frac{1}{1.354}$

@ 900K  $\frac{C_p}{R} = 1.121$   $\frac{C_v}{R} = 0.834$   $\frac{P}{T} = \frac{1}{1.344}$

$\frac{T_x - T_{min}}{T_{max} - T_{min}} = \frac{P_{max} - C_{min}}{C_{max} - C_{min}} \Rightarrow \frac{850 - 800}{900 - 800} = \frac{x - 1.099}{1.121 - 1.099}$

a.  $T_4$

$$\frac{T_1}{T_2} = \left(\frac{V_2}{V_1}\right)^{k-1} = \left(\frac{V_3}{V_4}\right)^{k-1} = \frac{T_4}{T_3} \quad \left| \begin{array}{l} V_2 = V_3 \\ V_4 = V_1 \end{array} \right.$$

$$r = 10 = \frac{V_{\max}}{V_{\min}} = \frac{V_1}{V_2} = \frac{V_1}{V_3} = \frac{V_4}{V_3} \quad \frac{10V_3 = V_1}{T_1 = T_4}$$

$$\frac{T_1}{T_2} = \left(\frac{V_2}{V_1}\right)^{k-1} \Rightarrow T_1 V_1^{k-1} = T_2 V_2^{k-1} \quad \left(\frac{V_3}{V_1}\right)^{10} = \frac{1}{10}$$

$$\left(\frac{4}{16}\right)^{5-1} \left(\frac{1}{4}\right)^4 = \frac{4^4}{2^4 \cdot 16} = \frac{256}{16} = 16$$

$$\left(\frac{4}{2}\right) \leq 16$$

Process 1-2: Polytropic compression

$$\Rightarrow T_2 = T_1 \left(\frac{V_1}{V_2}\right)^{k-1} = (60 + 273) [10]^{1.3-1} = 333 (10)^{0.3} = 664.4 \text{ K}$$

$$P_1 V_1^n = P_2 V_2^n$$

$$C = P_1 V_1^n = P_2 V_2^n$$

$$PV = mRT$$

$$= \frac{P_2 V_2 - P_1 V_1}{1-n}$$

$$P = \frac{mRT}{V} \quad \frac{mRT_2 - mRT_1}{1-n} = \frac{mR(T_2 - T_1)}{1-n}, n \neq 1 \text{ (KJ)}$$

$$W_{12} = \frac{R(T_2 - T_1)}{1-n} \text{ KJ/kg} = \frac{(0.287 \text{ KJ/kg} \cdot \text{K})(664.4 - 333) \text{ K}}{1-1.3} = 95.11 \text{ KJ/kg} = -317.03 \text{ KJ/kg}$$

-2-

Process 2-3: Constant volume heat addition

$$Pv = RT \Rightarrow \frac{P_2 V_2}{P_3 V_3} = \frac{RT_2}{RT_3} \quad (V_2 = V_3)$$

$$\frac{P_2}{P_3} = \frac{T_2}{T_3} \Rightarrow P_2 T_3 = T_2 P_3$$

$$T_3 = \left(\frac{P_3}{P_2}\right) T_2$$

$$C.M.E.P = \frac{W_{\text{net}}}{V_{\max} - V_{\min}} = \frac{W_{\text{net}}}{V_1 - V_2} = \frac{8000 \text{ kPa}}{1.99526 \text{ kPa}} = 4009.5 \text{ kPa}$$

$$\text{from } P_1 V_1 = RT_1$$

$$V_1 = \frac{RT_1}{P_1} = \frac{(0.287 \text{ KJ/kg} \cdot \text{K})(333 \text{ K})}{100 \text{ kPa}} = 2.663.9 \text{ K}$$

$$* Q_{in} = U_3 - U_2 = C_v(T_3 - T_2) = (0.823 \text{ KJ/kg} \cdot \text{K})(2663.9 - 664.4) \text{ K} = 1,645.58 \text{ KJ/kg}$$

Process 3-4: Constant volume Polytropic expansion:

$$T_4 = T_3 \left(\frac{V_3}{V_4}\right)^{n-1} = 2,663.9 \text{ K} \left(\frac{1}{10}\right)^{1.3-1}$$

$$P_3 V_3 = RT_3$$

$$P_4 V_4 = RT_4$$

$$\frac{P_3 V_3}{P_4 V_4} = 1$$

$$P_4 = P_3 \left(\frac{V_3}{V_4}\right)^n = (8000 \text{ kPa}) \left(\frac{1}{10}\right)^{1.3} = 400.95 \text{ kPa}$$

$$W_{34} = \frac{R(T_4 - T_3)}{1-n}$$

$$= \frac{(0.287 \text{ KJ/kg} \cdot \text{K})(1,335.1 - 2,663.9) \text{ K}}{1-1.3} = 1,271.22 \text{ KJ/kg}$$

$$P_3 V_3^n = P_4 V_4^n \quad P_4 = P_3 \left(\frac{V_3}{V_4}\right)^n$$

Process 4-1: Constant volume heat rejection

b.  $W_{\text{net,out}}, \eta_{\text{th}}$

$$W_{\text{net,out}} = W_{34} - W_{12}$$

$$= 1,271.22 \text{ KJ/kg} - 317.03 \text{ KJ/kg} = 954.19 \text{ KJ/kg} \approx 954.2 \text{ KJ/kg} \approx 58\%$$

$$\eta_{\text{th}} = \frac{W_{\text{net,out}}}{Q_{in}} = \frac{954.2 \text{ KJ/kg}}{1,645.58 \text{ KJ/kg}} = 0.58$$

MEP =  $\frac{W_{net}}{V_{max} - V_{min}} = \frac{W_{net}}{V_1 - V_2}$

From  $P_1 V_1 = RT_1$

$$V_1 = \frac{RT_1}{P_1} = \frac{(0.287 \frac{kJ}{kg \cdot K}) * 333K}{100 kPa}$$

$$V_1 = V_{max} = 0.9557 \frac{m^3}{kg}$$

&  $V_{min}$  from  $r = \frac{V_{max}}{V_{min}}$

$$V_{min} = \frac{V_{max}}{r} = \frac{0.9557 \frac{m^3}{kg}}{10} = 0.09557 \frac{m^3}{kg}$$

$$\Rightarrow MEP = \frac{954.19 \frac{kJ}{kg}}{(0.9557 - 0.09557) \frac{m^3}{kg}}$$

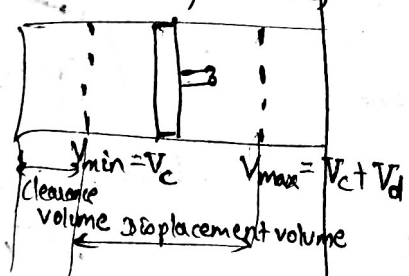
$$= 1,109.355 \frac{kJ}{m^3} = 1,109.355 \frac{N \cdot m}{m^3} = 1,109.355 kPa$$

d.  $\dot{n} = \frac{2 \dot{W}_{net}}{m_t \dot{W}_{net}}$  where  $\dot{W}_{net}$  = Net power output = 70 kW

$m_t$  = Total mass contained in the cylinder

$\dot{W}_{net}$  = Work net out

2  $\rightarrow$  rev/cycle (Four stroke)



From  $r = \frac{V_{max}}{V_{min}}$

$V_1$  = Total volume

$$V_1 = V_c + V_d$$

$$r = \frac{V_c + V_d}{V_c}$$

$$10 = \frac{V_c + (2.2 \times \frac{1000}{1000} m^3)}{V_c}$$

$$10V_c = V_c + 2.2$$

$$\Rightarrow 10V_c - V_c = 2.2 \Rightarrow 9V_c = 2.2$$

$$V_c = \frac{2.2}{9} m^3 = 0.000244 m^3$$

$$V_1 = 0.000244 m^3 + 0.0022 m^3 = 0.00244 m^3$$

$\Rightarrow m_t$  from  $P_1 V_1 = m_t R T_1$

$$m_t = \frac{P_1 V_1}{R T_1} = \frac{(100 kPa) * (0.00244 m^3)}{(0.287 \frac{kJ}{kg \cdot K}) * (333K)} = 0.00255 kg$$

-3-

$$\therefore \dot{n} = (2 \frac{rev}{cycle}) * \frac{50 \frac{kg}{s} * \frac{1 \frac{min}{60s}}{1 \frac{min}{60s}}}{(0.00255 \frac{kg}{kg}) * (954.19 \frac{kJ}{kg})}$$

$$= 2,465.9 \frac{rev}{min}$$

e. Specific fuel consumption,  $sfc$  ( $\frac{g}{kWh}$ )

$$sfc = \frac{m_f}{\frac{W_{net} * m_t}{P_{103/1p}}}$$

where  $m_f$  - mass of fuel consumed/burned during one cycle

$m_f$  from  $AF = \frac{m_a}{m_f} = \frac{m_t - m_f}{m_f}$

$$\Rightarrow 16 = \frac{(0.00255 kg) - m_f}{m_f}$$

$$16 m_f + m_f = 17 m_f = 0.00255 kg$$

$$m_f = 0.00015 kg$$

$$\therefore sfc = \frac{0.00015 kg}{954.19 \frac{kJ}{kg} * 0.00255 kg}$$

$$= 0.0000616476 \frac{kg}{kJ} * \frac{1000 \frac{g}{kg}}{1 \frac{min}{60s}} = 221.939 \frac{g}{kWh}$$

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00727106

MEP =  $\frac{W_{net}}{V_{max} - V_{min}} = \frac{W_{net}}{V_1 - V_2}$

From  $P_1 V_1 = RT_1$

$$V_1 = \frac{RT_1}{P_1} = \frac{(0.287 \text{ kJ/kg} \cdot \text{K}) * 333 \text{ K}}{100 \text{ kPa}}$$

$$V_1 = V_{max} = 0.9557 \text{ m}^3/\text{kg}$$

&  $V_{min}$  from  $r = \frac{V_{max}}{V_{min}}$

$$V_{min} = \frac{V_{max}}{r} = \frac{0.9557 \text{ m}^3/\text{kg}}{10} = 0.09557 \text{ m}^3/\text{kg}$$

$$\Rightarrow \text{MEP} = \frac{954.19 \text{ kJ/kg}}{(0.9557 - 0.09557) \text{ m}^3/\text{kg}}$$

$$= 1,109.355 \text{ kJ/m}^3$$

$$= 1,109.355 \text{ kPa}$$

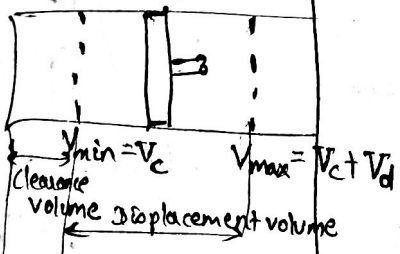
d.  $\dot{n} = \frac{2 \dot{W}_{net}}{m_t \dot{W}_{net}}$

where  $\dot{W}_{net}$  = Net power output = 70 kW

$m_t$  = Total mass contained in the cylinder

$\dot{W}_{net}$  = Work net out

2  $\rightarrow$  rev/cycle (Four stroke)



From  $r = \frac{V_{max}}{V_{min}}$

$V_1$  = Total volume

$$V_1 = V_c + V_d$$

$$r = \frac{V_c + V_d}{V_c}$$

$$10 = \frac{V_c + (2.2 \text{ kg} * \frac{1000 \text{ m}^3}{1000 \text{ kg}})}{V_c}$$

$$10V_c = V_c + 2.2$$

$$\Rightarrow 9V_c = 2.2 \Rightarrow V_c = 0.244 \text{ m}^3$$

$$V_c = 0.0022 \text{ m}^3 = 0.00024 \text{ m}^3$$

$$V_1 = 0.00024 \text{ m}^3 + 0.0022 \text{ m}^3 = 0.00244 \text{ m}^3$$

\*  $m_t$  from  $P_1 V_1 = m_t R T_1$

$$m_t = \frac{P_1 V_1}{R T_1} = \frac{(100 \text{ kPa}) * (0.00244 \text{ m}^3)}{(0.287 \text{ kJ/kg} \cdot \text{K}) * (333 \text{ K})} = 0.00255 \text{ kg}$$

-3-

$$\therefore \dot{n} = (2 \text{ rev/cycle}) * \frac{50 \text{ (rev/min)} * \frac{1 \text{ min}}{60 \text{ s}}}{(0.00255 \text{ kg}) * (954.19 \text{ kJ/kg})}$$

$$* \left( \frac{60 \text{ s}}{1 \text{ min}} \right)$$

$$= 2,465.9 \text{ rev/min}$$

e. Specific fuel consumption,  $sfc$  ( $\text{g/kWh}$ )

$$sfc = \frac{m_f}{\frac{W_{net} * m_t}{P_{103}/p}}$$

where  $m_f$  - mass of fuel consumed/burned during one cycle

$m_f$  from  $AF = \frac{m_a}{m_f} = \frac{m_t - m_f}{m_f}$

$$\Rightarrow 16 = \frac{(0.00255 \text{ kg}) - m_f}{m_f}$$

$$= \frac{m_f}{m_f}$$

$$16m_f + m_f = 17m_f = 0.00255 \text{ kg}$$

$$m_f = 0.00015 \text{ kg}$$

$$\therefore sfc = \frac{0.00015 \text{ kg}}{954.19 \text{ kJ} * 0.00255 \text{ kg}}$$

$$= 0.0000616476 \text{ kg/kJ}$$

$$\left( \frac{1 \text{ kJ}}{1 \text{ kWh}} \right) * \frac{3600 \text{ s}}{1 \text{ h}}$$

$$= 221.939 \text{ g/kWh}$$

-4-

- #2. #9-89. A six-cylinder, four stroke, 4.5 L compression-ignition engine operates on the ideal diesel cycle with a compression ratio of 17. The air is at 95 kPa & 55°C at the beginning of the compression process & the engine speed is 2000 rpm. The engine uses light diesel fuel with heating value of 42,500 kJ/kg, an air-fuel ratio of 24, & a combustion efficiency of 98 percent. Using constant specific heats at 850 K, determine:
- the max. temp. in the cycle & the thermal efficiency, cutoff ratio
  - the net work output per cycle & the thermal efficiency,
  - the mean effective pressure
  - the net power output, &
  - the specific fuel consumption, in g/kWh, defined as the ratio of the mass of the fuel consumed to the net work produced.

### Solution

Given  
6-cylinder / four stroke / 4.5 L / CI Engine  
 $r = 17$

$$P_1 = 95 \text{ kPa} = 0.0045 \text{ m}^3$$

$$T_1 = 55^\circ\text{C} + 273 = 328 \text{ K}$$

$$n = 2000 \text{ rpm}$$

$$\text{Fuel: } \text{GHV} = 42,500 \text{ kJ/kg}$$

$$\text{AF} = 24$$

$$\eta_{\text{combustion}} = 98\% = 0.98$$

$$C_p = 1.110 \text{ kJ/kg}\cdot\text{K}$$

$$C_v = 0.823 \text{ kJ/kg}\cdot\text{K}$$

$$\gamma = 1.349$$

$$R = 0.287 \text{ kJ/kg}\cdot\text{K}$$

### Required

$$a. T_{\text{max}} = T_3, \quad r_c = \frac{V_3}{V_2} = \frac{V_3}{V_2}$$

$$b. \text{Whe}_{\text{net}}^{\text{output}} / \text{cycle}, \quad \eta_{\text{th}}$$

$$c. \text{MEP}$$

$$d. \text{net power output} = \dot{W}_{\text{net}}$$

$$e. \text{sfc in g/kWh}$$

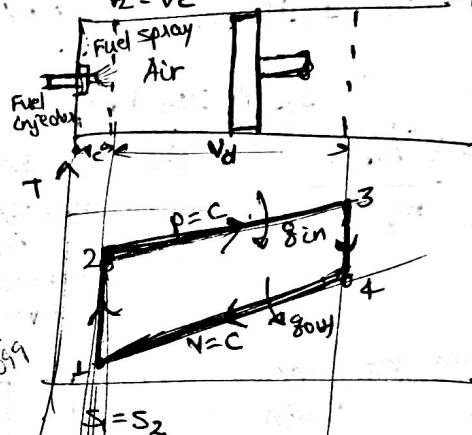
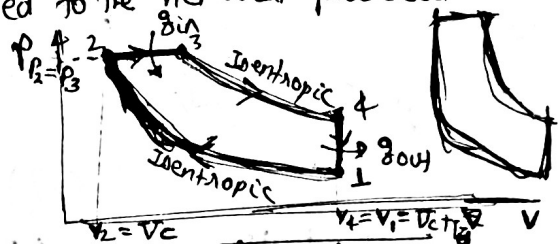
### Assumption

- Air standard assumptions are applicable.
- $\Delta KE$  &  $\Delta PE \approx 0$
- Air is an ideal gas with constant specific heats.

### Analysis

$$a. \dot{Q}_{\text{in}} = \dot{m} C_p (T_3 - T_2)$$

$$\dot{Q}_{\text{in}} = \dot{m} C_p (T_3 - T_2)$$



Process 1-2: Isentropic compression

$$T_2 = T_1 \left( \frac{V_1}{V_2} \right)^{\gamma-1} = (55+273) (17)^{1.349-1} = 881.658 \text{ K}$$

$$P_2 = P_1 \left( \frac{V_1}{V_2} \right)^{\gamma} = 95 \text{ kPa} (17)^{1.349} = 4,341.09 \text{ kPa}$$

$$\text{Compression} = r = 17 = \frac{V_1}{V_2} = \frac{V_{\text{max}}}{V_{\text{min}}}$$

Process 2-3: Constant pressure heat addition

$$\dot{Q}_{\text{in}} = \dot{m}_f \text{GHV} \eta_c$$

$$\dot{m}_f \text{ from } \text{AF} = \frac{m_a}{m_f} = \frac{m - m_f}{m_f}$$

→  $m$  Total mass contained in the cylinder

$$\text{From } P_1 V_1 = m R T_1$$

$$m = \frac{P_1 V_1}{R T_1}$$

$$V_1 = V_c + V_d$$

$$r = \frac{V_1}{V_c} = \frac{V_c + V_d}{V_c} \quad -5-$$

$$1.7 = \frac{V_c + 0.0045 \text{ m}^3}{V_c}$$

$$V_c = \frac{0.0045 \text{ m}^3}{16} = 0.00028125 \text{ m}^3$$

$$\Rightarrow V_1 = 0.00028125 \text{ m}^3 + 0.0045 \text{ m}^3 = 0.00478125 \text{ m}^3$$

$$\therefore m = \frac{9.5 \text{ kPa} \times 0.00478125 \text{ m}^3}{(0.287 \text{ kJ/kg} \cdot \text{K} \times 328 \text{ K})}$$

$$= 0.004825 \text{ kg}$$

$$\Rightarrow AF = 2.4 = \frac{(0.004825 \text{ kg}) - m_f}{m_f}$$

$$m_f = \frac{0.004825 \text{ kg}}{2.5}$$

$$= 0.00193 \text{ kg}$$

$$\therefore Q_{in} = 0.00193 \text{ kg} \times 42,500 \text{ kJ/kg} \times 0.98$$

$$= 6.93 \text{ kJ}$$

$$= 8.03845 \text{ kJ}$$

$$Q_{in} = m c_p (T_3 - T_2)$$

$$T_3 = \frac{Q_{in}}{m c_p} + T_2$$

$$= \frac{6.93 \text{ kJ}}{(0.004825 \text{ kg}) \times (1.11 \text{ kJ/kg} \cdot \text{K})} + (881.658 \text{ K})$$

$$= 2,175.59 \text{ K}$$

$$= 2,175.59 \text{ K}$$

$$\approx 2383 \text{ K}$$

$$r_c = \frac{V_3}{V_2} = \frac{T_3}{T_2} = \frac{2382.55}{881.658} = 2.7$$

$$\text{From } \frac{V_2}{V_3} = \frac{m R T_2}{m R T_3} \Rightarrow \frac{V_2}{V_3} = \frac{T_2}{T_3}$$

$$b. W_{net, out} = Q_{in} - Q_{out} \text{ for a cycle}$$

$$Q_{out} = m c_v (T_4 - T_1)$$

Process 4: 1 constant volume heat rejection  
 $c_v$  is used.

$$Q_{out} = m c_v (T_4 - T_1)$$

$$= (0.004825 \text{ kg}) \times (0.823 \text{ kJ/kg} \cdot \text{K}) \times$$

$T_4$  from process 3-4: Isentropic expansion

$$T_4 = T_3 \left( \frac{V_3}{V_4} \right)^{\gamma-1}$$

$$= 2,382.55 \text{ K}$$

$$\text{From } r_c = \frac{V_3}{V_2}$$

$$r = \frac{V_1}{V_2} \Rightarrow V_2 = \frac{V_1}{r} = \frac{0.00478125 \text{ m}^3}{1.7} = 0.0028125 \text{ m}^3$$

$$\text{And } V_3 = r_c V_2$$

$$= 2.7 \times 0.0028125 \text{ m}^3$$

$$= 0.00759375 \text{ m}^3$$

$$\Rightarrow T_4 = 2,382.55 \text{ K} \times \left( \frac{0.00759375}{0.00478125} \right)^{1.39-1} = 378.405 \text{ K}$$

$$\Rightarrow Q_{out} = (0.004825 \text{ kg}) \times (0.823 \text{ kJ/kg} \cdot \text{K}) \times (378.405 \text{ K} - 328 \text{ K}) = 0.215 \text{ kJ}$$

$$\therefore W_{net, out} = 8.03845 \text{ kJ} - 0.215 \text{ kJ} = 7.823 \text{ kJ}$$

$$\therefore \eta_{th} = \frac{W_{net, out}}{Q_{in}} = \frac{7.823 \text{ kJ}}{8.03845 \text{ kJ}} = 0.973 = 97.3\%$$

$$c. MEP = \frac{W_{net, out}}{V_1 - V_2} = \frac{7.823 \text{ kJ}}{(0.00478125 \text{ m}^3 - 0.00028125 \text{ m}^3)} = 968.88 \text{ kPa} \approx 96.9 \text{ kPa}$$

$$d. W_{net} = W_{net} \cdot \frac{n}{2}$$

$$= 4.36 \text{ kJ/cycle} \times \frac{1000}{2} \left( \frac{\text{rev/min}}{\text{cycle}} \right) \times \left( \frac{1 \text{ min}}{60 \text{ s}} \right)$$

$$= 72.66 \text{ kW}$$

$$\approx 72.7 \text{ kW}$$

$$e. \text{Sfc} = \frac{m_f}{W_{net}} = \frac{0.000193 \text{ kg}}{4.36 \text{ kJ}} \times \left( \frac{1000 \text{ J}}{1 \text{ kJ}} \right) \times \left( \frac{1 \text{ h}}{3600 \text{ s}} \right) \times \left( \frac{3600 \text{ s}}{1 \text{ h}} \right)$$

$$= 159.357 \text{ g/kWh}$$

$$\approx 159 \text{ g/kWh}$$

### #3 Dual combustion

Given

Ideal dual cycle

$$r = 18$$

$$r_c = 1.1$$

$$T_1 = 291 \text{ K}$$

$$P_1 = 90 \text{ kPa}$$

$$\text{At } 1000 \text{ temp: } Q = 1.005 \text{ kJ/kg} \cdot \text{K}$$

$$C_v = 0.718 \text{ kJ/kg} \cdot \text{K}$$

$$R = 0.287 \text{ kJ/kg} \cdot \text{K}$$

$$k = 1.4$$

Assumption

- The air standard assumptions are applicable
- $\Delta KE + \Delta PE$  are negligible
- Air is an ideal gas with constant specific heats

Analysis

Properties at various states:

$$T_2 = T_1 \left( \frac{V_1}{V_2} \right)^{k-1} = T_1 r^{k-1} = (291 \text{ K})^{1.4} = 877.4 \text{ K}$$

$$P_2 = P_1 \left( \frac{V_1}{V_2} \right)^k = P_1 r^k = (90 \text{ kPa})^{1.4} = 4343 \text{ kPa}$$

$$P_x = P_3 = r_c P_2 = (1.1)(4343 \text{ kPa}) = 4777 \text{ kPa}$$

$$T_x = \left( \frac{P_x}{P_2} \right) T_2 = \left( \frac{4777 \text{ kPa}}{4343 \text{ kPa}} \right) (877.4 \text{ K}) = 965.1 \text{ K}$$

$$r_c = \frac{V_3}{V_2} = \frac{P_2}{P_3}$$

$$V_3 = \frac{P_2}{P_3} V_2 = \frac{4343 \text{ kPa}}{4777 \text{ kPa}} V_2$$

$$V_3 = 0.91 \text{ V}_2$$

$$V_3 = 0.91 V_2$$

$$T_3 = T_x \left( \frac{V_3}{V_2} \right)^{k-1} = T_x r_c^{k-1} = 965.1 \text{ K} \cdot (0.91)^{1.4} = 1351 \text{ K}$$

$$T_4 = T_3 \left( \frac{V_3}{V_4} \right)^{k-1}$$

$$r_c = \frac{V_3}{V_4} = r \left( \frac{V_3}{V_4} \right)$$

$$\frac{V_3}{V_4} = \frac{r_c}{r}$$

Applying 1st law to each processes:

$$W_{1-2} = C_v (T_2 - T_1) = \dots = 46.7 \text{ kJ/kg}$$

$$Q_{2-x} = C_v (T_x - T_2) = \dots = 63.0 \text{ kJ/kg}$$

$$Q_{x-3} = C_p (T_3 - T_x) = \dots = 387.8 \text{ kJ/kg}$$

$$Q_{in} = W_b = C_p (T_3 - T_x)$$

$$W_{x-3} = Q_{x-3} - C_v (T_3 - T_x) = 387.8 \text{ kJ/kg}$$

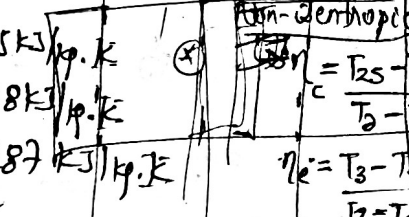
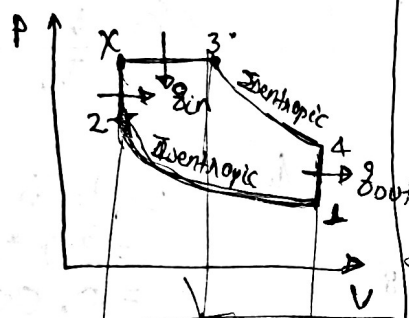
$$W_{3-4} = C_v (T_3 - T_4) = \dots = 594.4 \text{ kJ/kg}$$

$$W_{net} = W_{3-4} + W_{x-3} - W_{1-2} = 594.4 + 110.7 - 46.7 = 658.4 \text{ kJ/kg}$$

$$\eta_{th} = \frac{W_{net}}{Q_{in}} = \frac{658.4 \text{ kJ/kg}}{110.7 \text{ kJ/kg}} = 0.671$$

$$m = \frac{P_1 V_1}{R T_1} = \dots$$

$$\dot{W}_{net} = m \dot{W}_{net} \cdot \dot{n}$$



#4 #9-110

-7-

9-98 → 9-110A

9-108 - 9-122

0-135 → 9-

9-98

9-108

9-119

9-128

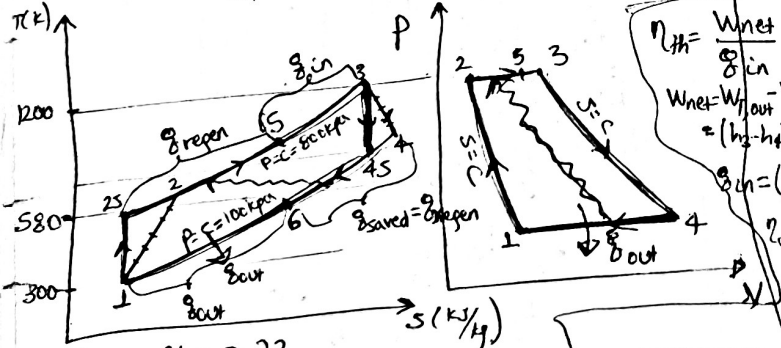
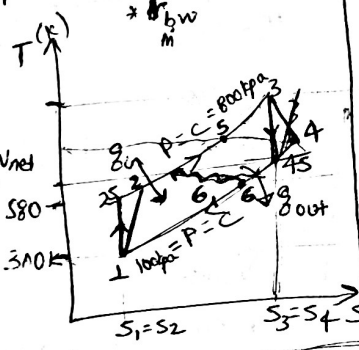
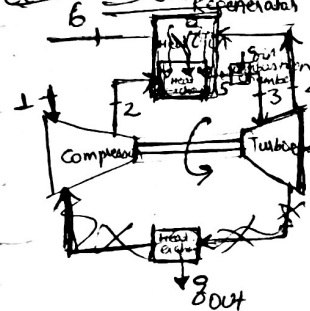
#9-98

#4. Air enters the compressor of a regenerative gas turbine engine @ 300K & 100kPa, where it is compressed to 800kPa & 580K. The regenerator has an effectiveness of 72%, & the air enters the turbine at 1200K. For a turbine efficiency of 86 percent, determine a) the amount of heat transfer in the regenerator &

b) the thermal efficiency.

Assume variable specific heats for air.

Given Solution



$$\epsilon = 72\% = 0.72$$

$$\eta_T = 86\% = 0.86$$

Required

a.  $\dot{Q}_{\text{regen}}$ b.  $\eta_{\text{th}}$ 

$$h_4 = 1277.79 \text{ kJ/kg} - [0.86 (1277.79 \text{ kJ/kg} - 719.75 \text{ kJ/kg})]$$

$$= 797.87 \text{ kJ/kg}$$

$$\dot{Q}_{\text{regen}} = \epsilon (h_4 - h_2)$$

$$= 0.72 (797.87 \text{ kJ/kg} - 586.04 \text{ kJ/kg})$$

$$= 152.52 \text{ kJ/kg}$$

Assumption

- The air standard assumptions are applicable.
- Air is an ideal gas with variable specific heats.
- $\Delta KE$  &  $\Delta PE$  are negligible.

$$sp. \text{ heat} = f(T)$$

$$\hookrightarrow T_r, V_r$$

Analysis

$$\dot{Q}_{\text{regen, act}} = h_5 - h_2$$

$$\dot{Q}_{\text{regen, max}} = h_5 - h_2 = h_4 - h_2$$

$$\epsilon = \frac{\dot{Q}_{\text{regen, act}}}{\dot{Q}_{\text{regen, max}}} = \frac{h_5 - h_2}{h_4 - h_2} \approx \frac{T_5 - T_2}{T_4 - T_2}$$

$$\dot{Q}_{\text{regen}} = \epsilon (h_4 - h_2)$$

 $h_4 = ?$ 

$$\eta_T = \frac{h_3 - h_4}{h_3 - h_{4s}}$$

$$h_4 = h_3 - \eta_T (h_3 - h_{4s})$$

$$h_4 = 1277.79 \text{ kJ/kg} - 0.86 (1277.79 \text{ kJ/kg} - 719.75 \text{ kJ/kg})$$

$$h_4 = 797.87 \text{ kJ/kg}$$

$$r_p = \frac{P_2}{P_1} = \frac{800 \text{ kPa}}{100 \text{ kPa}} = 8$$

$P_r$  is relative pressure (unitless) used for calculation of isentropic process.

- Used to evaluate property changes of ideal gases during isentropic process since it accounts for the variation of sp. heats w temp.

$$T_2 \xrightarrow{\text{read}} P_{r2} = \frac{P_2}{P_1} P_{r1}$$

$$T_1 \xrightarrow{\text{read}} P_{r1}$$

$$P_{r3} @ 1200 \text{ K}; P_{r3} = 238.0$$

$$P_{r4} = \frac{P_4}{P_3} P_{r3} = \left(\frac{1}{8}\right) (238.0) = 29.75$$

$$P_{r4} = 29.75 \rightarrow h = 713.27$$

$$P_{r4} = 29.75 \rightarrow ?$$

$$P_{r4} = 30.38 \rightarrow h = 724.04$$

$$P_{r4} - P_{r4 \text{ min}} = ? - h_{\text{min}}$$

$$P_{r4 \text{ max}} - P_{r4 \text{ min}} \quad h_{\text{max}} - h_{\text{min}}$$

$$P_{r4} - 28.80 = ? - 713.27$$

$$30.38 - 28.80 \quad 724.04 - 713.27$$

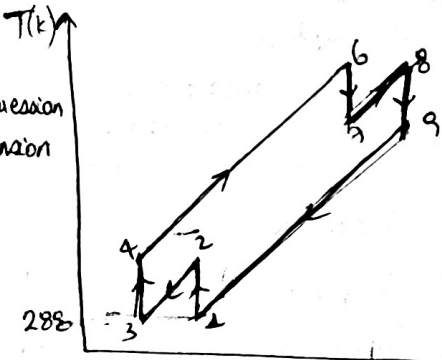
$$1.58 \quad (10.77)$$

$$10.77 P_{r4} - 310.176 = 1.58 ?$$

$$h_{4s} = 719.75 \text{ kJ/kg}$$

Ideal regenerative cycle w  
 - 2 stages of compression  
 - 2 " " expansion

(78)



$\eta_{th}$  w regenerator:-?

Assumption

- The air stan
- Air is ideal gas w constant sp. heats @ room temp.
- $\Delta KE$  &  $\Delta PE \approx 0$

Analysis

Expressions for the isentropic compression & expansion processes are

$$T_2 = T_1 (r_p)^{(k-1)/k}$$

$$T_4 = T_3 \left( \frac{1}{r_p} \right)^{(k-1)/k}$$

For ideal regenerator:

$$T_5 = T_4$$

$$T_6 = T_2$$

$$\eta_{th} = 1 - \frac{q_{out}}{q_{in}} = 1 - \frac{T_6 - T_1}{T_3 - T_5}$$

$$= 1 - T_1 \left( \frac{T_6}{T_1} - 1 \right) / \left( T_3 \left( 1 - \frac{T_5}{T_3} \right) \right)$$

$$\frac{T_3}{T_4} = \left( P_r \right)^{(k-1)/k}$$

$$= 1 - \frac{T_1}{T_3} \left( \frac{T_2}{T_1} - 1 \right) / \left( 1 - \frac{T_4}{T_3} \right)$$

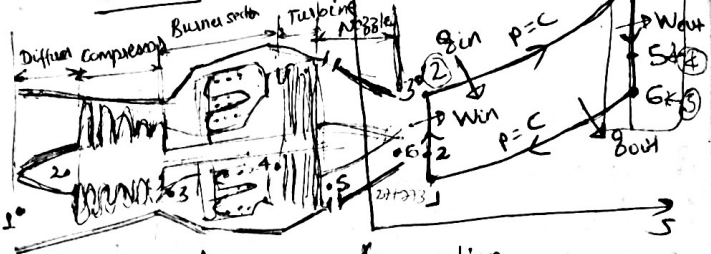
$$= 1 - \frac{T_1}{T_3} \left( r_p^{(k-1)/k} - 1 \right) / \left( 1 - r_p^{-(k-1)/k} \right)$$

$$= 1 - \frac{T_1}{T_3} r_p^{(k-1)/k} \left( \frac{1 - \frac{1}{r_p^{(k-1)/k}}}{1 - r_p^{-(k-1)/k}} \right)$$

#5 / 9 = 119 /

Consider an aircraft powered by a turbojet engine that has a pressure ratio of 12. The aircraft is stationary on the ground, held in position by its brakes. The ambient air is at 27°C & 95 kPa & enters the engine at a rate of 10 kg/s. The jet fuel has a heating value of 42,700 kJ/kg, and it is burned completely at a rate of 0.2 kg/s. Neglecting the effect of the diffuser & disregarding the slight increase in mass at the engine exit as well as the inefficiencies of engine components, determine the force that must be applied on the brakes to hold the plane stationary.

Solution  
 Given



$$r_p = 12 = \frac{P_2}{P_1}$$

$$T_a = 27^\circ C$$

$$P_a = 95 kPa$$

$$\dot{m} = 10 kg/s$$

$$HV = 42,700 kJ/kg$$

$$\dot{m}_f = 0.2 kg/s$$

Assumption

- Steady operating conditions exist
- The air standard assumptions are applicable
- Air is an ideal gas w variable sp. heats.
- $\Delta KE$ ,  $\Delta PE \approx 0$ , except at the nozzle exit.

Analysis

$$\text{Brake force} = \text{Thrust} = \dot{m} (V_{exit} - V_{inlet})$$

$$V_{exit} = V_6$$

$$V_{inlet} = 0$$

To find  $V_6$

$$\dot{E}_{in} - \dot{E}_{out} = \Delta \dot{E}_{system}$$

$$\dot{E}_{in} = \dot{E}_{out}$$

$$\dot{m} h_5 + \frac{\dot{m} V_5^2}{2} = \dot{m} h_6 + \frac{\dot{m} V_6^2}{2}$$

$$h_5 + \frac{V_5^2}{2} = h_6 + \frac{V_6^2}{2}$$

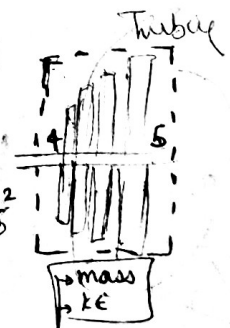
$$0 = h_5 - h_6 + \frac{V_5^2 - V_6^2}{2}$$

$$V_6^2 = 2(h_5 - h_6)$$

$$V_6 = \sqrt{2(h_5 - h_6)}$$

$$\left[ \begin{matrix} h_4 \\ h_6 \end{matrix} \right] \rightarrow ? \quad h_5 \rightarrow ? \text{ Variable sp. heats}$$

$$P_{15} = P_{13} \left( \frac{P_5}{P_3} \right)$$



# Compressor

$$T_1 = 27 + 273 = 300 \text{ K} \rightarrow h_1$$

$$\frac{P_2}{P_1} = 12$$

Variable specific heats of air.

$$\Rightarrow P_{r3} = \frac{P_3}{P_1} P_{r1}$$

$$@ T_1 = 300 \text{ K} \rightarrow h_1 = 300.19 \text{ kJ/kg}$$

$$P_{r1} = 1.3860$$

$$\Rightarrow P_{r3} = (12) * 1.3860$$

$$= 16.632$$

$$h = 607.02 \quad P_r = 16.28$$

$$h = ? \quad P_r = 16.632$$

$$h = 628.07 \quad P_r = 18.36$$

$$\frac{h_3 - 607.02}{628.07 - 607.02} = \frac{16.632 - 16.28}{18.36 - 16.28}$$

$$h_3 = 607.02 + \frac{(628.07 - 607.02) * (16.632 - 16.28)}{(18.36 - 16.28)}$$

$$= 610.58 \text{ kJ/kg}$$

$$Q_{in} = \dot{m}_{fuel} * HV$$

$$= (0.2 \text{ kg/s}) * (42,700 \text{ kJ/kg})$$

$$= 8,540 \text{ kJ/s}$$

$$\dot{q}_{in} = \frac{Q_{in}}{\dot{m}} = \frac{8,540 \text{ kJ/s}}{10 \text{ kg/s}} = 854 \text{ kJ/kg}$$

$$\Rightarrow \dot{q}_{in} = h_4 - h_3$$

$$\Rightarrow h_4 = h_3 + \dot{q}_{in}$$

$$= 610.58 \text{ kJ/kg} + 854 \text{ kJ/kg}$$

$$= 1,464.58 \text{ kJ/kg}$$

$$375.3 \quad 1443.60$$

$$424.2 \quad 1491.44$$

# Turbine

Ideally

$$W_{comp, in} = W_{turb, out}$$

$$h_3 - h_1 = h_4 - h_5$$

$$h_5 = h_4 - (h_3 - h_1)$$

$$= 1464.58 \text{ kJ/kg} - (610.58 \text{ kJ/kg} - 300.19 \text{ kJ/kg})$$

$$= 1,154.19 \text{ kJ/kg}$$

Nozzle:

$$P_{r6} = P_{r4} \left( \frac{P_6}{P_4} \right)$$

$$= 396.7 \text{ kJ/kg} \left( \frac{1}{12} \right)$$

$$= 33.058$$

Enthalpy

$$h = 734.82 \quad 32.02$$

$$? \quad 33.058$$

$$745.62 \quad 33.72$$

$$\frac{? - 734.82}{745.62 - 734.82} = \frac{33.058 - 32.02}{33.72 - 32.02}$$

$$h_6 = 741.4 \text{ kJ/kg}$$

$$V_6 = \sqrt{2(h_4 - h_6)}$$

$$= \sqrt{2(1,154.19 \text{ kJ/kg} - 741.4 \text{ kJ/kg})}$$

$$= 908.6 \text{ m/s}$$

$$\text{Brake force} = \text{Thrust} = \dot{m}(V_{exit} - V_{inlet})$$

$$= 10 \text{ kg/s} * 908.6 \text{ m/s}$$

$$= 9086 \text{ N}$$

$$\frac{? - 375.3}{(424.2 - 375.3)} = \frac{1464.58 - 1443.6}{(1491.44 - 1443.6)} \Rightarrow P_{r4} = 396.7 \text{ kJ/kg}$$

#9-128

-10-

Determine the exergy destruction associated with each of the processes of the Brayton cycle described in problem 9-98, assuming a source temp. of 1260 K and a sink temp. of 300 K. Also, determine the exergy of the exhaust gases at the exit of the regenerator.

Take  $P_{\text{exhaust}} = P_0 = 100 \text{ kPa}$

From problem 9-98

|                    |         |
|--------------------|---------|
| $g_{\text{in}} =$  | $S_1^0$ |
| $g_{\text{out}} =$ | $S_6^0$ |
| $T_1 =$            | $S_1^0$ |
| $h_2 =$            | $S_2$   |
| $T_3 =$            | $S_3^0$ |
| $h_4 =$            | $S_4^0$ |
| $h_5 =$            | $S_5^0$ |

$$\begin{aligned}
 X_{\text{destroyed}, 61} &= T_0 S_{\text{gen}, 61} \\
 &= T_0 (S_1 - S_6 + \frac{\dot{Q}_{R, 61}}{T_R}) \\
 &= T_0 (S_1^0 - S_6^0 - R \ln \frac{P_1}{P_0} + \frac{\dot{Q}_{R, 61}}{T_R}) \\
 &= \underline{\hspace{2cm}}
 \end{aligned}$$

Nothing that  $h_0 = h_{@T_0} = \underline{\hspace{2cm}}$

Stream exergy at exit of the regenerator

$$\begin{aligned}
 \phi_6 &= (h_6 - h_0) - T_0 (S_6 - S_0) + \frac{V_6^2}{2} + g z_6^0 \\
 \text{where} \\
 S_6 - S_0 &= S_6 - S_1 = S_6^0 - S_1^0 - R \ln \frac{P_0}{P_1} = 0
 \end{aligned}$$

Energy balance on the heat exchanger =  $\underline{\hspace{2cm}}$

$$h_5 - h_2 = h_4 - h_6 \quad \phi_6 = \underline{\hspace{2cm}}$$

$$h_6 = \underline{\hspace{2cm}}$$

$$S_6^0 = \underline{\hspace{2cm}}$$

$$X_{\text{destroyed}, 12} = T_0 S_{\text{gen}, 12} = T_0 (S_2 - S_1) = T_0 \left[ S_2^0 - S_1^0 - R \ln \frac{P_2}{P_0} \right] \quad \text{1/rp}$$

$$X_{\text{destroyed}, 34} = T_0 S_{\text{gen}, 34} = T_0 (S_4 - S_3) = T_0 \left[ S_4^0 - S_3^0 - R \ln \frac{P_4}{P_3} \right] \quad \text{1/rp}$$

$$\begin{aligned}
 X_{\text{destroyed}, \text{regen}} &= T_0 S_{\text{gen}, \text{regen}} = T_0 \left[ (S_5 - S_2) + (S_6 - S_4) \right] \\
 &= T_0 \left[ (S_5^0 - S_2^0) + (S_6^0 - S_4^0) \right]
 \end{aligned}$$

$$\begin{aligned}
 X_{\text{destroyed}, 53} &= T_0 S_{\text{gen}, 53} = T_0 \left[ S_3 - S_5 - \frac{\dot{Q}_{R, 53}}{T_R} \right] \\
 &= T_0 \left[ S_3^0 - S_5^0 - R \ln \frac{P_3}{P_5} - \frac{\dot{Q}_{R, 53}}{T_R} \right] \\
 &= \underline{\hspace{2cm}}
 \end{aligned}$$